SEQUENTIAL INFORMATION CASCADE WITH MARTINGALE FAIRNESS

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1 Introduction

I am going to describe an interesting game. Call it the Mystery Guest game.There are N number of players. Define $[N] = [1, N] \cap \mathbb{N}$. Let $i \in [N]$ be the i^{th} player. The game starts from $i = 1$ and goes to $i = N$. Every person is allowed to ask a question and then given a chance to answer the question, which is some historical person. If they answer correctly, they win and the game ends, if they answer wrong they lose money (points) , and if they pass they lose nothing and the game moves to the next player. Let M be the number of cycles for which the game keeps on going, and L be the number of time a player is allowed to answer a question. Obviosuly $L <= M$. Asking a question gives the next person more information to answer. We can represent the increase in probability as a Bayesian updation in probability. Let Φ_{ij} be the probability of the player i getting the answer correct during the cycle j. Let p_{ij} be the number of points gained by person i during cycle j, and n_{ij} be the points lost on answering wrong. Let l_i be the number of guesses that player i can make.
The expected points of player i during cycle j is:

$$E_{ij} = \Phi_{ij} p_{ij} - (1 - \Phi_{ij}) n_{ij} \tag{1}$$

There are many ways to describe fairness. I am using the Invariance of the Expectation E_{ij} as the condition of fairness. Thus

$$\boldsymbol{\forall i, i^{'} \in [N], j, j^{'} \in [M], E_{ij} = E_{i^{'}j^{'}} = C} \tag{2}$$

Call C the fairness constant. Now let's say for the first player $p_{11} = P_0 > 0$ and $n_{11} = 0$.

$$\boldsymbol{E_{11} = \Phi_{11} P_0 = C} \tag{3}$$

I want to enforce another condition where $p_{ij} = P_0 - x_{ij}, x_{11} = 0$ and $n_{ij} = x_{ij}$, basically a simple additive and subtractive evolution of points(which was the case in the original game). In this constrained framework

$$\begin{aligned} \boldsymbol{E_{ij}} &\boldsymbol{= \Phi_{ij}(P_0 - x_{ij}) - (1 - \Phi_{ij}) x_{ij}} && (4) \\ &\boldsymbol{= \Phi_{ij} P_0 - \Phi_{ij} x_{ij} - x_{ij} + \Phi_{ij} x_{ij}} && (5) \\ &\boldsymbol{= \Phi_{ij} P_0 - x_{ij} = \Phi_{11} P_0} && (6) \\ \boldsymbol{\Rightarrow x_{ij}} &\boldsymbol{= P_0(\Phi_{ij} - \Phi_{11})} && (7) \end{aligned}$$

Imagine you are a player playing this game. How many times and when will you choose to answer to ensure you win this game(your chances of winning the game is maximum)?. You want to answer before anyone else does correcly, and that implies that you will get more points than anyone else(everyone else is atmost 0 points and lower if they lost points). So maybe we should look at when is the best time to answer?
Let us begin with defining the game a bit more clearly. I need a way to signal if a player is going to answer a question or not in the future. Let Λ_{ij} be the probability that player i will answer during cycle j. $1 - \Lambda_{ij}$ is the probability of leaving the question. $\Lambda_{ij}\Phi_{ij}$ is the probability of answering correctly. Also the probability of the game ending at that specific turn. $\Lambda_{ij}(1 - \Phi_{ij})$ the probability of getting it wrong. The probability that the game will end at turn $i^{'}, j^{'}$, from the starting of the game is :

$$\prod_{i,j \in [i^{'}-1] \times [j^{'}-1]} ((1 - \Lambda_{ij}) + (\Lambda_{ij}\Phi_{ij}))(\Lambda_{i^{'}j^{'}}\Phi_{i^{'}j^{'}}) \tag{8}$$

The probability that the game will not end and player i will get another chance to play(player's chance will come back after a cycle(assuming $j < M$):

$$S_{ij} = \prod_{k \in [N]/\{i\}} (\textbf{player k either loses or passes}) \tag{9}$$

$$= \prod_{k \in [i+1,N] \cap \mathbb{N}} ((1-\Lambda_{kj}) + \Lambda_{kj}(1-\Phi_{kj})) \cdot \prod_{k \in [1,i-1] \cap \mathbb{N}} ((1-\Lambda_{k(j+1)}) + \Lambda_{k(j+1)}(1-\Phi_{k(j+1)})) \tag{10}$$

Here we are assuming that $\forall i \in [N], l_i >= M - j + 1$, given $j < M$. This is quite a unique situation and we should call this probability the full capacity Survival probability. We can define a more constrained probability of survival $S^{'}_{ij}$. It is very easy to see that $S_{ij} <= S_{ij}(A)$.

Alternately we can define an indicator function to make this look simpler. First, from (9)

$$S_{ij} = \prod_{k \in [i+1,N] \cap \mathbb{N}} (1 - \lambda_{kj}\Phi_{kj}) \cdot \prod_{k \in [1,i-1] \cap \mathbb{N}} (1 - \lambda_{k(j+1)}\Phi_{k(j+1)}) \tag{11}$$

Define

$$\mathbb{I}_k = \{1, \textbf{ if } l_k > 0; 0, \textbf{ if } l_k = 0\} \tag{12}$$

Using the indicator,

$$S^{'}_{ij} = \prod_{k \in [i+1,N] \cap \mathbb{N}} (1 - \lambda_{kj}\Phi_{kj}\mathbb{I}_k) \cdot \prod_{k \in [1,i-1] \cap \mathbb{N}} (1 - \lambda_{k(j+1)}\Phi_{k(j+1)}\mathbb{I}_k) \tag{13}$$

The survival probability will be useful when we'll construct the Value function.

2 How to find the optimal strategy for player i, assuming perfect rationality?

With perfect rationality, every player knows everything about every other player. In this case what is the optimal way to play this game ?? We can use backward induction to find the optimal. Define a Value function $V_i(j, l_i)$, which represents the maximum expected value for i from cycle j and onwards. At the immediate point the player can choose to answer $\Lambda_{ij} = 1$. Now if the player gets it right then the expected point is C, and if they answer wrong with probability $1 - \Phi_{ij}$, the player's guess decreases by 1, and the game must not end before cycle $j + 1$. The survival probability gives the chances of this. Define

$$V_{ans} = C + (1 - \Phi_{ij})S^{'}_{ij}V_i(j + 1, l_i - 1) \tag{14}$$

This is the Value of answering. If player passes then they cannot get C and the value depends on the future value

$$V_{pass} = S^{'}_{ij}V_i(j + 1, l_i) \tag{15}$$

Now player will choose between passing and answering, depending on whichever has a higher value of expectation

$$V_i(j, l_i) = \max(V_{ans}, V_{pass}) \tag{16}$$

If

- $V_{ans} >= V_{pass}$, answering is the optimal choice
- $V_{pass} > V_{ans}$, passing is the optimal choice

Backward Induction

At the terminal state where $i = N, j = M$,

$$V_N(M, l_N) = \max(V_{ans}, V_{pass}) = \max(C, 0) = C \tag{17}$$

(assuming $l_N > 0$. For player $N - 1$, we already know that player N will answer, therefore player $N - 1$ will choose knowing this information.

$$V_{N-1}(M, L_{N-1}) = \max(V_{ans}, V_{pass}) = \max(C, 0) = C \tag{18}$$

Infact for $j = M, \forall i \in [N], V_{ans} = C, V_{pass} = 0$ (assuming $l_i > 0$). Another interesting thing For all players in the last cycle, if they have a guess left, they should guess because

$$\forall i \in [N], V_i(M, l_N) = C \tag{19}$$

Now moving to the previous cycle $M-1$.

$$V_N(M-1, L_N) = \max(C + (1 - \Phi_{N(M-1)})S^{'}_{N(M-1)}V_N(M, L_N - 1), S^{'}_{N(M-1)}V_N(M, L_N)) \tag{20}$$

Survival Probability of player N at cycle $M-1$

$$S^{'}_{N(M-1)} = \prod_{k=1}^{N-1}(1 - \Lambda_{k(j+1)}\Phi_{k(j+1)}\mathbb{I}_k) \tag{21}$$

Since we already know that $\Lambda_{iM} = \Lambda^{*}_{iM} = 1$,

$$S^{'}_{N(M-1)} = \prod_{k=1}^{N-1}(1 - \Phi_{k(M)}\mathbb{I}_k) \tag{22}$$

Let us divide this situation into two cases.
Case I:

$$l_N = 1$$

This case means that either player N can choose to guess right now or later but cannot guess in both the cases.

$$V_i(j, l_i) = 0, \ \textbf{if } l_i <= 0$$

The Value function in this case is

$$V_N(M-1, l_N) = \max(C + (1 - \Phi_{N(M-1)})S^{'}_{N(M-1)} \cdot 0, S^{'}_{N(M-1)} \cdot V_N(M, 1)) \tag{23}$$

$$= \max(C, S^{'}_{N(M-1)}C). \tag{24}$$

Since $S^{'}_{N(M-1)} \leq 1 \Rightarrow C \geq S^{'}_{N(M-1)}C$ In this case N should choose to guess since $V_{ans} \geq V_{pass}$.
Case II:

$$l_N > 1$$

Then,

$$V_N(M-1, l_N) = \max(C + (1 - \Phi_{N(M-1)})S^{'}_{N(M-1)}C, S^{'}_{N(M-1)}C) \qquad \textbf{(From (19))}$$

Now from there the player will guess if

$$V_{ans} >= V_{pass}$$

$$\Rightarrow 1 + (1 - \Phi_{N(M-1)})S^{'}_{N(M-1)} >= S^{'}_{N(M-1)}$$

If $\Phi_{N(M-1)} < 1$, then

$$1 + (1 - \Phi_{N(M-1)})S^{'}_{N(M-1)} >= S^{'}_{N(M-1)} \tag{25}$$

$$S^{'}_{N(M-1)} <= \frac{1}{\Phi_{N(M-1)}} \tag{26}$$

But $\frac{1}{\Phi_{N(M-1)}} >= 1$ and maximal value of $S^{'}_{N(M-1)}$ is 1.

$$S^{'}_{N(M-1)} =< 1 \tag{27}$$

If $\Phi_{N(M-1)} = 0$,

$$S^{'}_{N(M-1)} =< 1 \tag{28}$$

The player will pass if $V_{ans} < V_{pass}$

$$S'_{N(M-1)} > 1 \tag{29}$$

which is not possible. Therefore a player should always choose to guess in both the cases. Player N should take a guess no matter what.

3 Proving that under strict Martingale fairness and risk neutral agents, waiting is economically irrational

There is one condition under which for every player, immediately guessing is the optimal choice.

$$S_{ij}^{'} \leq \frac{1}{1 + \Phi_{ij}(l_{ij} - 1)} \tag{30}$$

The above inequality is the required condition. We will now use Induction to show that under this condition, all players, whatever cycle it may be, will choose to guess $\Lambda_{ij}^{*} = 1 \forall i, j \in [N] \times [M]$ or $V_{ij} = V_{ij}^{ans} \geq V_{ij}^{pass}$
The base case is already derived by us

$$V_{iM} = V_{iM}^{ans} = C$$

The inductive hypothesis is

$$V_{i(j+1)} = V_{i(j+1)}^{ans} = C + (1 - \Phi_{i(j+1)})S_{i(j+1)}^{'}V_{i(j+2)} \tag{31}$$

We can first show these important bounds which are true $\forall i, j \in [N] \times [M]$

$$V_i(j, l_i) - V_i(j, l_i - 1) \leq C \tag{32}$$

$$V_i(j, l_i) \leq Cl_i \tag{33}$$

Let us look at V_{ij} now

$$V_{ij} = \begin{cases} C + (1 - \Phi_{ij})S_{ij}^{'}V_i(j+1, l_i - 1), & \Lambda_{ij}^{*} = 1 \\ S_{ij}^{*}V_i(j+1, l_i), & \Lambda_{ij}^{*} = 0 \end{cases} \tag{34}$$

$$\begin{aligned}
&V_{ij}^{ans} \geq V_{ij}^{pass} \\
&\Rightarrow C + (1 - \Phi_{ij})S_{ij}^{'}V_i(j+1, l_i - 1) \geq S_{ij}^{*}V_i(j+1, l_i) \\
&\Rightarrow C - \Phi_{ij}S_{ij}^{'}V_i(j+1, l_i - 1) \geq S_{ij}^{'}(V_i(j+1, l_i) - V_i(j+1, l_i - 1)) \\
&\Rightarrow C - \Phi_{ij}S_{ij}^{'}V_i(j+1, l_i - 1) \geq S_{ij}^{'}C && \textbf{(From (32))} \\
&\Rightarrow C - C(l_i - 1)\Phi_{ij}S_{ij}^{'} \geq S_{ij}^{'}C && \textbf{(From(33))}
\end{aligned}$$

$$\Rightarrow S_{ij}^{'} \leq \frac{1}{1 + \Phi_{ij}(l_i - 1)} \tag{35}$$

With this condition we have proved that whenever $\boldsymbol{V_i(j+1, l_i) = V_{i(j+1)}^{ans}}$, it implies that $\boldsymbol{V_i(j, l_i) = V_{ij}^{ans}}$. Thus, at every turn, the optimal strategy is to take a guess assuming (35).
Let us prove that this is a subgame-perfect Nash Equilibrium. Assume that some player i during some cycle j chooses to pass their turn. We will show that expected points will be lower for this player(assuming (30)). So for player i during j

$$V_i^{'}(j, l_i) = V_{ij}^{pass} = S_{ij}^{'}V_i(j+1, l_i) \tag{36}$$

Compare it with $V_i(j, l_i)$

$$\begin{aligned}
V_i(j, l_i) - V_i^{'}(j, l_i) &= C + (1 - \Phi_{ij})S_{ij}^{'}V_i(j+1, l_i - 1) - S_{ij}^{'}V_i(j+1, l_i) \\
&= C + S_{ij}^{'}(V_i(j+1, l_i - 1) - V_i(j+1, l_i)) - \Phi_{ij}S_{ij}^{'}V_i(j+1, l_i - 1) \\
&\geq C - S_{ij}^{'}C - \Phi_{ij}S_{ij}^{'}C(l_i - 1) \\
&= C(1 - S_{ij}^{'}[1 + \Phi_{ij}(l_i - 1)]) \geq 0
\end{aligned}$$

Thus $\Lambda_{ij} = 0$ is never better than $\Lambda_{ij} = 1$. Obviously we are imposing a condition that the probability to survive is low enough to force players to guess. Or they must have enough confidence in answering (Φ_{ij}) and have enough guesses left. What if they have only one guess left ?? Then

$$S_{ij}^{'} \leq \frac{C}{V_i(j, l_i)} \tag{37}$$

And if the player cannot guess at all ($\Phi_{ij} = 0$)

$$S'_{ij} \leq \frac{C}{V_i(j, l_i) - V_i(j, l_i - 1)} \tag{38}$$

What about when a player is fully sure of the answer ($\Phi_{ij} = 1$)??

$$S'_{ij} \leq \frac{C}{V_i(j, l_i)}$$

4 Is the game still fair ?? Computing the overall expected points of the players

Player i gets L guesses to answer and the Nash Equilibrium shows that they will answer at every turn. The game ends at cycle $L \leq M$. So the player has L chances. Let Ω_m be the probability of getting to cycle m from cycle 1 for player i. What is the Expected number of points that a player playing this game can get? First let us compute the Expected points for a single cycle m.

$$\mathbb{E}_m = (1 - \Omega_m)0 + \Omega_m E_m = \Omega_m C \tag{39}$$

We can add this to find the total Expectation for player i, $\mathbb{G}$

$$\mathbb{G}(i) = \sum_{m=1}^{L} \mathbb{E}_m = \sum_{m=1}^{L} \Omega_m C \tag{40}$$

$$= C \sum_{m=1}^{L} \left[\prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] \tag{41}$$

It is easy to see why $\forall i \in \{1, 2, 3, N-1\}$

$$\mathbb{G}(i) \geq \mathbb{G}(i+1) \tag{42}$$

Note that

$$\begin{aligned}
\mathbb{G}(i+1) &= C \sum_{m=1}^{L} \left[\prod_{k=1}^{(m-1)N+(i+1)-1} (1 - \Phi_k) \right] \\
&= C \sum_{m=1}^{L} \left[(1 - \Phi_{((m-1)N+i)}) \prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] \\
&\leq C \sum_{m=1}^{L} \left[\prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] = \mathbb{G}(i)
\end{aligned}$$

Therefore the subsequent players never have a higher expectation. And if $0 < \Phi_{ij} < 1$, then the expectation keeps getting lower. Thus, it is always better to be a lower-number player; it is always better to start the game earlier. Now, the next thing I want to do is to know if there is a way to force the game to be globally fair as well.

5 Constant Payoffs and Changing Expectations

I will now relax the Martingale fairness condition. Assume

$$\forall i, j \in [N] \times [M], p_{ij} = p, n_{ij} = q \tag{43}$$

SO constant payoffs. The Expected points while answering is

$$E_{ij} = \Phi_{ij} p + (1 - \Phi_{ij}) q \tag{44}$$

The value function then will be ,

$$V_i(j, l_i) = \max \begin{cases} E_{ij} + (1 - \Phi_{ij}) S'_{ij} V_i(j+1, l_i - 1), & \Lambda_{ij} = 1 \\ S'_{ij} V_i(j+1, l_i), & \Lambda_{ij} = 0 \end{cases} \tag{45}$$

The Bellman Inequality is

$$\begin{aligned}
V_i^{ans}(j,l_i) &\geq V_i^{pass}(j,l)\\
E_{ij} + (1-\Phi_{ij})S'_{ij}V_i(j+1,l_i-1) &\geq S'_{ij}V_i(j+1,l_i)\\
\Phi_{ij}p - (1-\Phi_{ij})q + (1-\Phi_{ij})S'_{ij}V_i(j+1,l_i-1) &\geq S'_{ij}V_i(j+1,l_i)\\
\Phi_{ij} &\geq \frac{S'_{ij}(V_i(j+1,l_i)-V_i(j+1,l_i-1))+q}{p+q-S'_{ij}V_i(j+1,l_i-1)}
\end{aligned}$$

If i knows the probabilities of the other players winning then they should only guess if their own probability of answering is big enough.

$$\Rightarrow S'_{ij} \leq \frac{\Phi_{ij}p-(1-\Phi_{ij})q}{V_i(j+1,l_i)-(1-\Phi_{ij})V_i(j+1,l_i-1)} = \frac{E_{ij}}{V_i(j+1,l_i)-(1-\Phi_{ij})V_i(j+1,l_i-1)} \tag{46}$$

Obviously a player wont know about future probability of answering unless new information is revealed. But by Backward induction we can again show that under (46), the SPNE is to always guess.